

DURHAM 2 SSW, NH, USA

WMO: 726100

Lat: 43.1092N

Lon: 70.9484W

Elev: 63

StdP: 14.66

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 54795

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.1	4.1	-12.1	2.9	4.5	-7.9	3.6	8.9	12.6	20.0	11.5	19.2	2.2	N/A	0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	88.3	73.1	85.4	71.1	82.4	69.7	75.4	84.3	73.9	82.2	72.3	79.6	3.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.6	121.0	80.3	71.2	115.4	78.1	69.8	109.9	76.3	38.8	84.5	37.4	82.5	36.0	79.3	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
10.9	9.0	7.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-9.1	93.5	5.6	2.1	-13.1	95.0	-16.4	96.2	-19.6	97.4	-23.7	99.0
			-10.0	78.5	4.1	1.2	-13.0	79.4	-15.4	80.1	-17.7	80.8	-20.6	81.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.2	23.8	25.9	33.7	45.2	55.0	63.5	70.0	68.4	61.1	49.8	39.3	29.5	(6)
(7)		DBStd	17.85	10.62	9.70	9.21	8.32	7.53	6.73	5.15	5.47	7.00	7.65	8.20	8.86	(7)
(8)		HDD50	3295	813	675	512	192	30	1	0	2	100	335	635		(8)
(9)		HDD65	6910	1278	1094	970	596	324	107	13	27	156	474	772	1099	(9)
(10)		CDD50	2275	1	0	6	48	186	404	620	569	336	92	12	1	(10)
(11)		CDD65	420	0	0	0	2	14	61	168	132	41	2	0	0	(11)
(12)		CDH74	4059	0	0	8	74	251	638	1539	1154	374	21	0	0	(12)
(13)		CDH80	1162	0	0	1	22	70	180	470	322	96	1	0	0	(13)
(14)	Wind	WSAvg	2.8	3.2	3.5	3.8	3.7	2.9	2.3	2.0	1.9	2.0	2.5	2.7	2.9	(14)
(15)	Precipitation	PrecAvg	40.6	3.0	2.9	3.1	3.7	3.3	3.2	3.0	3.1	3.3	3.9	4.2	3.8	(15)
(16)		PrecMax	58.8	7.8	10.3	9.2	11.5	9.3	11.3	7.9	8.8	9.3	13.9	11.3	9.3	(16)
(17)		PrecMin	24.6	0.5	0.0	0.9	0.6	0.7	0.8	0.6	0.8	0.3	0.2	0.6	0.5	(17)
(18)		PrecStd	7.4	1.7	1.6	1.5	2.1	1.8	2.0	1.7	1.6	1.8	2.8	2.1	2.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.3	58.9	70.5	83.1	87.1	90.2	91.5	91.4	89.0	77.0	68.0	58.2	(19)
(20)		0.4%	MCWB	54.9	50.0	56.6	63.1	68.1	72.3	75.3	76.3	74.2	66.4	59.7	54.3	(20)
(21)		2%	DB	48.5	50.3	59.1	73.3	81.0	85.2	87.9	87.0	82.5	71.8	62.4	51.1	(21)
(22)		2%	MCWB	43.7	42.9	49.8	59.8	65.1	70.6	73.6	73.5	70.9	63.5	55.1	47.5	(22)
(23)		5%	DB	42.6	44.1	52.4	66.7	75.5	81.0	85.3	83.5	78.2	67.0	57.6	46.9	(23)
(24)		5%	MCWB	37.7	38.4	43.5	56.3	62.4	67.7	71.6	70.5	68.1	61.2	50.1	43.6	(24)
(25)		10%	DB	38.1	39.3	47.0	61.1	70.3	77.3	82.3	80.4	74.3	62.6	53.2	42.3	(25)
(26)		10%	MCWB	34.2	34.3	40.5	49.8	60.6	65.0	70.5	68.8	66.7	57.4	47.8	38.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.8	55.2	59.1	64.7	71.3	75.0	78.0	76.9	75.7	69.6	61.6	54.4	(27)
(28)		0.4%	MCDWB	57.5	59.4	72.9	72.0	83.8	84.8	88.3	87.3	84.2	73.8	66.8	55.8	(28)
(29)		2%	WB	45.1	46.1	53.2	61.2	68.3	72.9	76.1	74.6	73.0	65.4	56.3	48.9	(29)
(30)		2%	MCDWB	47.2	51.6	60.2	71.2	77.9	82.6	85.6	82.6	80.5	68.6	60.4	50.5	(30)
(31)		5%	WB	39.0	40.2	45.3	56.6	65.4	70.4	74.5	72.8	71.1	62.3	51.7	44.9	(31)
(32)		5%	MCDWB	42.0	45.7	50.7	64.3	73.5	77.8	82.9	79.6	77.1	66.6	56.3	47.3	(32)
(33)		10%	WB	35.0	36.7	40.9	51.3	61.9	67.9	72.8	71.0	68.7	59.0	48.1	40.1	(33)
(34)		10%	MCDWB	38.0	40.4	46.1	58.5	69.7	74.9	80.3	76.9	73.1	62.9	51.9	42.9	(34)
(35)	Mean Daily Temperature Range		MDBR	16.8	18.4	18.5	21.0	20.8	20.6	20.9	21.6	21.8	19.6	18.0	15.9	(35)
(36)		5% DB	MCDDBR	20.8	24.6	26.4	32.3	29.8	26.2	25.0	25.2	26.1	24.7	25.1	21.3	(36)
(37)			MCWBR	18.9	17.7	17.4	17.4	15.1	13.3	12.0	12.3	14.1	15.5	18.7	17.2	(37)
(38)			MCDDBR	20.6	24.5	25.7	25.3	26.0	23.3	21.6	20.5	21.3	20.4	22.6	19.2	(38)
(39)		5% WB	MCWBR	19.0	18.9	18.2	16.7	15.0	13.1	11.6	11.2	12.8	14.7	18.4	17.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.286	0.297	0.321	0.358	0.395	0.441	0.461	0.428	0.374	0.334	0.312	0.292	(40)	
(41)		taud	2.470	2.454	2.471	2.415	2.350	2.251	2.204	2.284	2.416	2.530	2.550	2.538	(41)	
(42)		Ebn at Noon	272	288	293	288	279	265	258	263	271	271	261	261	(42)	
(43)		Edh at Noon	23	28	31	35	39	43	44	40	32	25	21	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	542	817	1154	1461	1627	1752	1844	1671	1345	871	584	437	(44)	
(45)		RadStd	36	64	75	158	186	156	120	81	82	79	51	48	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (75 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+149	+59	(47)

Nomenclature: See separate page